

BUSINESS INTELLIGENCE PORTFOLIO PROJECT

Credit Utilization & Customer Usage Analysis

EVALUATING CREDIT UTILIZATION EFFICIENCY AND
REVENUE OPTIMIZATION OPPORTUNITIES

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Executive Summary

This analysis evaluates whether portfolio growth is constrained by credit availability or customer spending behavior.

Using 641,914 transaction records across 5,000 customers, the study finds that average credit utilization remains relatively low despite substantial transaction volume.

The analysis suggests that the largest growth opportunity lies in improving customer utilization behavior rather than expanding credit limits. The findings reveal a structural monetization gap driven by low utilization despite substantial available credit capacity.



Key Findings

- Avg Utilization = 22.87%
- Majority customers are Low Usage
- Credit Limit $\neq$ Spending Growth
- Large unused credit capacity exists

Business Context & Problem Statement

BUSINESS CONTEXT

Credit portfolios are commonly evaluated using:

- Total Spend
- Customer Growth
- Credit Limit Allocation

However, these metrics may conceal inefficiencies in capital productivity and credit allocation efficiency.

PROBLEM STATEMENT

Despite generating 86.38M total spend, average utilization remains only 22.87%.

Key Questions:

1. Are customers fully utilizing available credit?
2. Which customer segments drive utilization?
3. Is portfolio growth constrained by credit supply or spending behavior?

Analysis Scope & KPI Baseline

WHY THESE METRICS MATTER

- Customers → Portfolio scale
- Spend → Revenue generation
- Utilization → Capital productivity
- Transactions → Usage intensity

5K

Total Customers

86.38M

Total Spend

22.87%

Avg Utilization

127.86

Avg Transactions

Data & Methodology

DATASET

- 641,914 Transactions
- 5,000 Customers
- 23 Variables

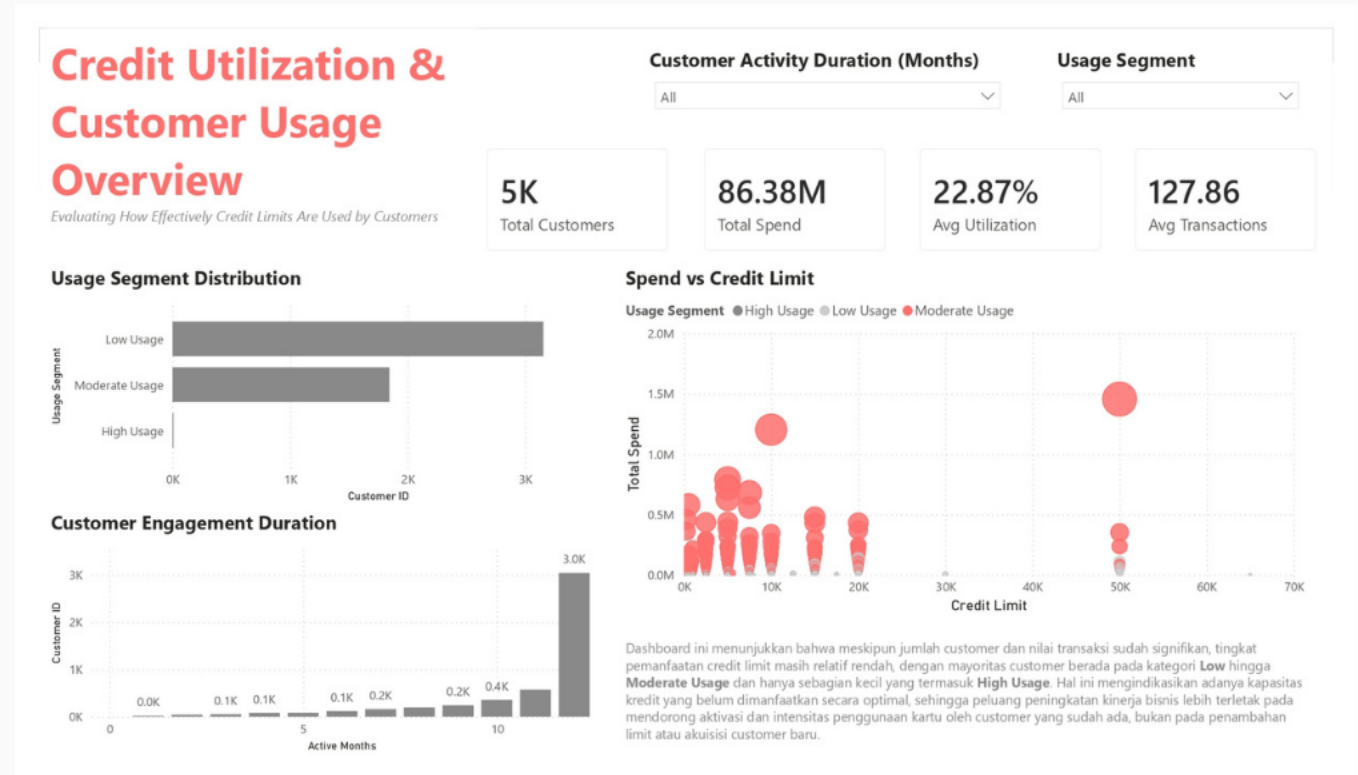
METHODOLOGY

- Data Cleaning (Python)
- Customer-Level Aggregation (SQL)
- Dashboard Modeling (Power BI)

WORKFLOW

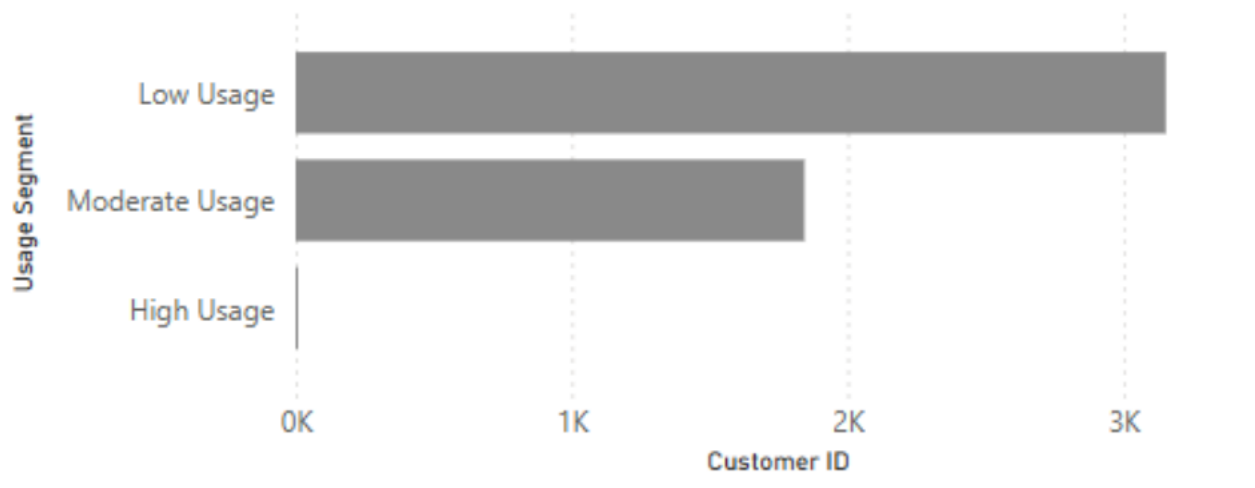
Raw Data → Preparation → Analysis → Dashboard → Recommendation

Dashboard Overview



Utilization Segment Analysis

Usage Segment Distribution



Findings

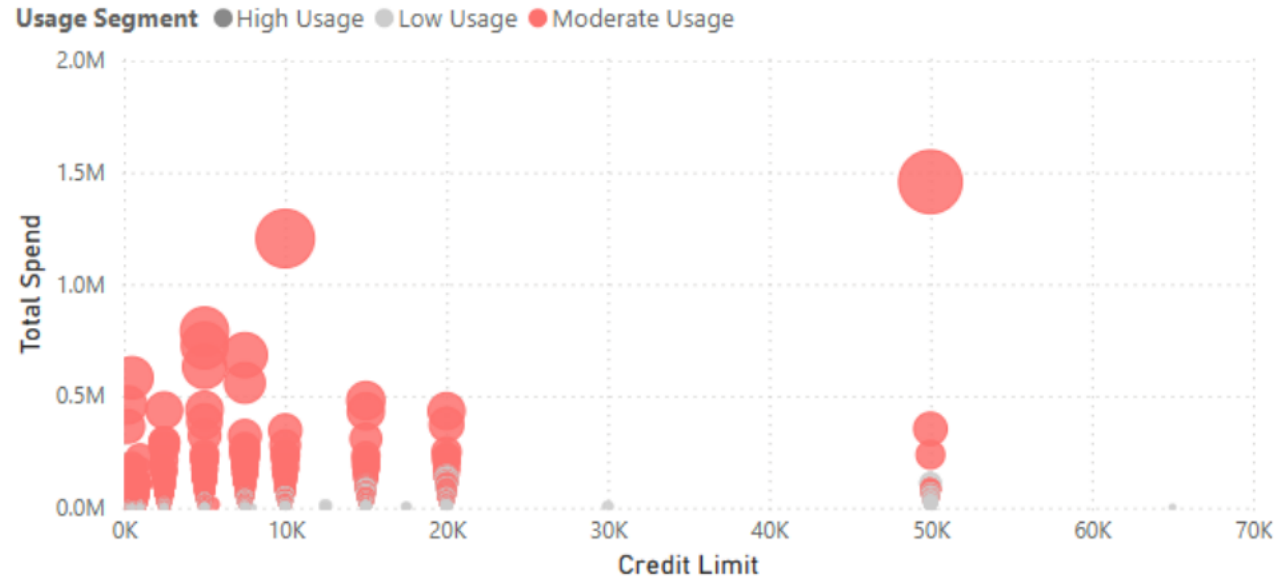
- Low Usage dominates portfolio
- Moderate Usage contributes second-largest share
- High Usage customers represent only a small portion
- Revenue generation is likely concentrated within a relatively small High Usage segment, creating potential dependency risk and reducing portfolio balance.

Business Interpretation

A significant portion of available credit remains idle, indicating that capital deployment efficiency is below potential despite a sizeable customer base. Ins underutilized.

Credit Limit vs Spend Analysis

Spend vs Credit Limit



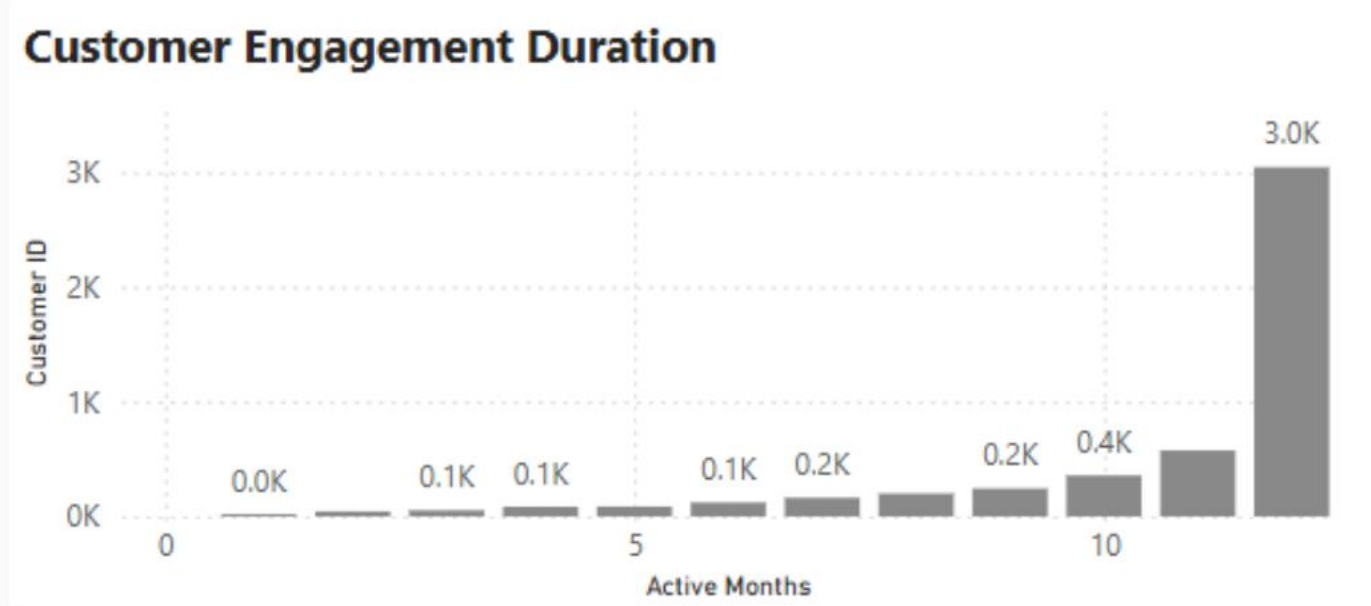
Findings

The analysis suggests that increasing credit limits alone is unlikely to drive proportional spending growth. Portfolio expansion should prioritize customer activation and spending behavior rather than credit capacity expansion.

Business Interpretation

- Increasing limits alone may not generate proportional revenue growth.
- Behavioral activation appears more impactful than credit expansion.

Customer Engagement Analysis



Findings

Customer retention is not the primary business challenge. Existing customers remain active, suggesting that monetization opportunities lie in increasing usage frequency and spending intensity.

Business Interpretation

- The portfolio does not appear to suffer from retention issues.
- The primary challenge lies in transaction intensity and utilization behavior.

Root Cause Analysis

CORE BUSINESS INSIGHT

The analysis indicates that portfolio growth is not constrained by customer volume, retention, or credit availability.

Instead, the primary limitation stems from low utilization among existing customers despite sufficient available credit and sustained engagement.

5 WHYS ANALYSIS:

WHY IS PORTFOLIO UTILIZATION LOW?

Why 1: Most customers belong to the Low Usage segment.

Why 2: A large portion of available credit remains unused.

Why 3: Customer spending growth lags behind credit allocation growth.

Why 4: Customer engagement with credit usage remains limited.

Why 5: Portfolio strategies emphasize credit allocation more than utilization activation.

Strategic Recommendations

IMMEDIATE ACTIONS

- Cashback campaigns
- Spending incentives
- Utilization activation programs

MEDIUM-TERM ACTIONS

- Segment-based marketing initiatives
- Personalized customer offers
- Usage reactivation campaigns

LONG-TERM ACTIONS

- Utilization prediction model
- Dynamic credit management framework
- Next Best Offer recommendation engine

KPI Monitoring Framework

Portfolio Performance Targets

KPI	Current	Target
Credit Utilization Rate	22.87%	30%
Total Spend	86.38M	+15–20%
Low Usage Share	High	Reduce
Moderate Usage Share	Moderate	Increase

Ongoing Monitoring Areas

- Utilization Trend
- Segment Migration
- Customer Activation Rate
- Spending Growth
- Transaction Frequency

Future Monitoring Opportunities

- Customer Lifetime Value (CLV)
- Utilization Prediction
- Revenue Contribution by Segment

Strategic focus should shift from transaction volume toward utilization efficiency, customer activation, and capital productivity.

Conclusion

KEY TAKEAWAY

Although the portfolio serves more than 5,000 customers and generates substantial transaction volume, average credit utilization remains relatively low.

Most customers are not fully utilizing their available credit capacity, while higher credit limits do not consistently translate into increased spending.

Future portfolio growth should therefore focus on improving customer utilization behavior, increasing engagement, and maximizing capital productivity rather than broadly expanding credit limits.



Thank You